

BEC303 Network Analysis — Module 3 Quiz

Transient Behavior and Initial Conditions

Field	Value
Subject code	BEC303
Module	3 — Transient Behavior and Initial Conditions
Duration	45 minutes
Maximum marks	50 (Part A 20 + Part B 30)
Attempts allowed	1
Passing marks	20
Textbook	W. H. Hayt, J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, 8th ed., 2012

Instructions

- Part A carries 20 one-mark multiple choice questions. Exactly one option is correct.
- Part B carries 13 short questions totalling 30 marks.
- State the continuity rule you are invoking whenever you write a value at $t = 0+$.
- Time constants must be computed from the resistance seen after switching.
- Non-programmable scientific calculators are permitted.

Course outcome and Bloom coverage

Band	Questions	Share	Target
L1–L2 Remember / Understand	11	33%	~30%
L3–L4 Apply / Analyse	19	58%	~45%
L5–L6 Evaluate / Create	3	9%	~25%

Course outcome spread: CO3: 33. Total questions: 33.

Part A — Multiple choice (20 × 1 mark = 20 marks)

A1. Which quantity is guaranteed continuous across a switching instant? [CO3 | L1 | 1 mark | easy | Ch.7, pp.217-225]

- (A) Capacitor voltage
- (B) Capacitor current
- (C) Resistor voltage
- (D) Inductor voltage

A2. In DC steady state an inductor behaves as [CO3 | L1 | 1 mark | easy | Ch.7, pp.225-235]

- (A) a short circuit
- (B) an open circuit

- (C) a resistance equal to ωL
- (D) a voltage source

A3. An uncharged capacitor at $t = 0+$ is correctly represented as [C03 | L2 | 1 mark | medium | Ch.7, pp.217-247]

- (A) a short circuit
- (B) an open circuit
- (C) a 1 V source
- (D) a resistance of $1/\omega C$

A4. A 2 H inductor carrying 3 A is switched onto a 100 Ω resistor. The resistor voltage immediately after switching is [C03 | L3 | 1 mark | medium | Ch.7, pp.225-235]

- (A) 300 V
- (B) 0 V
- (C) 6 V
- (D) 150 V

A5. The time constant of a source-free RL circuit is [C03 | L1 | 1 mark | easy | Ch.8, pp.261-268]

- (A) L/R
- (B) R/L
- (C) RL
- (D) $1/(RL)$

A6. After one time constant an exponential decay has fallen to approximately [C03 | L2 | 1 mark | easy | Ch.8, pp.268-272]

- (A) 36.8% of its initial value
- (B) 50% of its initial value
- (C) 63.2% of its initial value
- (D) 13.5% of its initial value

A7. A 10 μF capacitor discharging through 1 k Ω has a time constant of [C03 | L2 | 1 mark | easy | Ch.8, pp.272-275]

- (A) 10 ms
- (B) 100 ms
- (C) 1 ms
- (D) 10 s

A8. The total energy dissipated when a charged capacitor discharges fully through R is [C03 | L4 | 1 mark | hard | Ch.8, pp.272-275]

- (A) $\frac{1}{2}Cv(0)^2$, independent of R
- (B) $\frac{1}{2}Cv(0)^2/R$
- (C) proportional to R
- (D) zero, since the capacitor is lossless

A9. A rectangular pulse of width T is most usefully written as [C03 | L2 | 1 mark | medium | Ch.8, pp.282-286]

- (A) $u(t) - u(t - T)$
- (B) $u(t) + u(t - T)$
- (C) $T \cdot u(t)$

- (D) $u(t) \cdot u(t - T)$

A10. The general first-order response formula is [C03 | L2 | 1 mark | medium | Ch.8, pp.286-295]

- (A) $x(t) = x(\infty) + [x(0+) - x(\infty)] \cdot e^{(-t/\tau)}$
- (B) $x(t) = x(0+) \cdot e^{(-t/\tau)}$
- (C) $x(t) = x(\infty) \cdot e^{(-t/\tau)}$
- (D) $x(t) = x(0+) + x(\infty) \cdot e^{(-t/\tau)}$

A11. A 20 V source is switched onto a series 5 Ω and 2 H with zero initial current. The current at $t = 0.4$ s is [C03 | L3 | 1 mark | medium | Ch.8, pp.286-295]

- (A) 2.53 A
- (B) 4 A
- (C) 1.47 A
- (D) 0 A

A12. For a 5 V supply and a 2.5 V threshold, the RC delay to reach threshold is [C03 | L4 | 1 mark | hard | Ch.8, pp.295-300]

- (A) $0.693 \cdot RC$
- (B) $1 \cdot RC$
- (C) $2.3 \cdot RC$
- (D) $0.5 \cdot RC$

A13. An RC circuit with $\tau = 1$ ms charges toward 10 V for 2 ms and is then disconnected. The discharge starts from [C03 | L4 | 1 mark | hard | Ch.8, pp.300-306]

- (A) 8.65 V
- (B) 10 V
- (C) 6.32 V
- (D) 3.68 V

A14. For a parallel RLC circuit the neper frequency is [C03 | L2 | 1 mark | medium | Ch.9, pp.321-334]

- (A) $\alpha = 1/(2RC)$
- (B) $\alpha = R/(2L)$
- (C) $\alpha = 1/\sqrt{LC}$
- (D) $\alpha = 2RC$

A15. A circuit with $\alpha = 1000 \text{ s}^{-1}$ and $\omega_0 = 10000 \text{ rad/s}$ is [C03 | L3 | 1 mark | medium | Ch.9, pp.321-334]

- (A) underdamped
- (B) overdamped
- (C) critically damped
- (D) unstable

A16. The critically damped solution form is [C03 | L3 | 1 mark | medium | Ch.9, pp.334-345]

- (A) $(A_1 + A_2 t) \cdot e^{(-\alpha t)}$
- (B) $A \cdot e^{(-\alpha t)}$
- (C) $A_1 e^{(s_1 t)} + A_2 e^{(s_2 t)}$ with distinct real s
- (D) $e^{(-\alpha t)}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$

A17. For a series RLC circuit, increasing R [C03 | L4 | 1 mark | hard | Ch.9, pp.345-359]

- (A) increases the damping
- (B) decreases the damping
- (C) leaves the damping unchanged
- (D) changes ω_0 but not α

A18. How many independent initial conditions are needed to determine a second-order response? [C03 | L2 | 1 mark | medium | Ch.9, pp.345-359]

- (A) Two
- (B) One
- (C) Three
- (D) Four

A19. When an inductive circuit is switched onto an AC supply, the largest transient offset occurs when switching at [C03 | L5 | 1 mark | hard | Ch.9, pp.359-361]

- (A) a voltage zero crossing
- (B) a voltage peak
- (C) a current peak
- (D) any instant, since the offset is fixed

A20. The final condition of a circuit driven by a sinusoidal source is [C03 | L3 | 1 mark | medium | Ch.10, pp.371-406]

- (A) a steady sinusoid found by phasor analysis
- (B) zero
- (C) a constant DC value
- (D) undefined

Part B — Descriptive (13 questions, 30 marks)

B1. State the two continuity rules for switching and justify each from the corresponding element law. [C03 | L2 | 3 marks | Ch.7, pp.217-235]

B2. Describe the complete procedure for evaluating initial conditions at $t = 0+$ in a second-order circuit. [C03 | L3 | 3 marks | Ch.8, pp.261-268]

B3. A 20 V source with 5 Ω and a 2 H inductor has been closed for a long time. At $t = 0$ a 15 Ω resistor is switched into the loop. Find $i(0+)$, $v_L(0+)$ and $di/dt(0+)$. [C03 | L3 | 3 marks | Ch.8, pp.261-268]

B4. A parallel RLC has $R = 500 \Omega$, $L = 10 \text{ mH}$ and $C = 1 \mu\text{F}$. Classify the damping and find the damped natural frequency. [C03 | L4 | 3 marks | Ch.9, pp.321-334]

B5. Explain why the critically damped response requires a $t \cdot e^{(-\alpha t)}$ term. [C03 | L4 | 2 marks | Ch.9, pp.334-345]

B6. Contrast the effect of increasing R in a series RLC with its effect in a parallel RLC. [C03 | L4 | 2 marks | Ch.9, pp.345-359]

B7. A 10 μF capacitor charged to 5 V discharges through 1 k Ω . Find τ , the initial current, and the total energy dissipated. [C03 | L3 | 2 marks | Ch.8, pp.272-275]

B8. Explain how to read a time constant from an oscilloscope trace by two different methods. [C03 | L3 | 2 marks | Ch.8, pp.268-272]

B9. A 5 V pulse of width 3 ms is applied to an initially uncharged RC circuit with $\tau = 1$ ms. Sketch and describe the capacitor voltage. [CO3 | L3 | 2 marks | Ch.8, pp.282-286]

B10. Design an RC power-ON delay of 100 ms for a 5 V supply with a 2.5 V logic threshold, using a 10 μ F capacitor. [CO3 | L5 | 2 marks | Ch.8, pp.295-300]

B11. Explain why the final value of a circuit driven by a sinusoid is not zero, and how it is found. [CO3 | L3 | 2 marks | Ch.10, pp.371-406]

B12. Explain why a 20 V source switched onto a series 5 Ω and 2 H gives $i(t) = 4(1 - e^{(-2.5t)})$ A, identifying each part of the response. [CO3 | L4 | 2 marks | Ch.8, pp.286-295]

B13. Explain why switching an inductive load onto an AC supply at a voltage zero produces the worst transient. [CO3 | L5 | 2 marks | Ch.9, pp.359-361]

Answer key — Part A

Q	Answer	CO	Bloom	Reference
A1	A	CO3	L1	Ch.7, pp.217-225
A2	A	CO3	L1	Ch.7, pp.225-235
A3	A	CO3	L2	Ch.7, pp.217-247
A4	A	CO3	L3	Ch.7, pp.225-235
A5	A	CO3	L1	Ch.8, pp.261-268
A6	A	CO3	L2	Ch.8, pp.268-272
A7	A	CO3	L2	Ch.8, pp.272-275
A8	A	CO3	L4	Ch.8, pp.272-275
A9	A	CO3	L2	Ch.8, pp.282-286
A10	A	CO3	L2	Ch.8, pp.286-295
A11	A	CO3	L3	Ch.8, pp.286-295
A12	A	CO3	L4	Ch.8, pp.295-300
A13	A	CO3	L4	Ch.8, pp.300-306
A14	A	CO3	L2	Ch.9, pp.321-334
A15	A	CO3	L3	Ch.9, pp.321-334
A16	A	CO3	L3	Ch.9, pp.334-345
A17	A	CO3	L4	Ch.9, pp.345-359
A18	A	CO3	L2	Ch.9, pp.345-359
A19	A	CO3	L5	Ch.9, pp.359-361
A20	A	CO3	L3	Ch.10, pp.371-406

Part A — why each option is right or wrong

A1. Correct answer: (A) Capacitor voltage

- (A) Correct — a jump in capacitor voltage would require infinite current.
- (B) Wrong - capacitor current is free to step and usually does at a switching instant.
- (C) Wrong - resistor voltage follows the current instantaneously and can jump.
- (D) Wrong - inductor voltage is the element that jumps; it is the current that is continuous.

A2. Correct answer: (A) a short circuit

- (A) Correct — di/dt is zero, so $v = L \cdot di/dt = 0$ and the element is a short.
- (B) Wrong - that is the DC behaviour of a capacitor.
- (C) Wrong - ω is zero in DC, so this reactance would be zero, which is a short, but the reasoning as stated is wrong.
- (D) Wrong - an inductor stores energy but does not impose a voltage in steady state.

A3. Correct answer: (A) a short circuit

- (A) Correct — with $V_0 = 0$ the initial-condition source has zero value, which is a short.
- (B) Wrong - an open circuit is the $t = \text{infinity}$ DC representation, not the $t = 0+$ one.
- (C) Wrong - the initial-condition source equals the actual initial voltage, which here is zero.
- (D) Wrong - this is a steady-state AC reactance and has no meaning at a switching instant.

A4. Correct answer: (A) 300 V

- (A) Correct — the current is continuous at 3 A, so $v = 3 \times 100 = 300$ V.
- (B) Wrong - this assumes the current starts from zero, which violates inductor continuity.
- (C) Wrong - this multiplies the inductance by the current, which has units of flux rather than volts.
- (D) Wrong - halving has no basis; the full continuous current flows through the resistor.

A5. Correct answer: (A) L/R

- (A) Correct — L/R has units of seconds and sets the decay rate.
- (B) Wrong - R/L is the neper frequency α for a series circuit, the reciprocal of the time constant.
- (C) Wrong - the product has the wrong units entirely.
- (D) Wrong - this is dimensionally incorrect and inverts the dependence on both elements.

A6. Correct answer: (A) 36.8% of its initial value

- (A) Correct — $e^{-1} = 0.368$.
- (B) Wrong - 50 percent corresponds to 0.693 time constants, not one.
- (C) Wrong - 63.2 percent is how far a rising response has travelled, not how much of a decay remains.
- (D) Wrong - 13.5 percent corresponds to two time constants.

A7. Correct answer: (A) 10 ms

- (A) Correct — $\tau = RC = 1000 \times 10e-6 = 0.01$ s.
- (B) Wrong - this is off by a factor of ten in the capacitance.
- (C) Wrong - this is off by a factor of ten in the other direction.
- (D) Wrong - this is off by three orders of magnitude, typically from a microfarad-to-farad slip.

A8. Correct answer: (A) $\frac{1}{2}Cv(0)^2$, independent of R

- (A) Correct — all the stored energy ends up in the resistor regardless of its value.
- (B) Wrong - dividing stored energy by resistance does not even have units of energy.

- (C) Wrong - R controls how fast the energy leaves, never how much.
- (D) Wrong - the capacitor is lossless but the resistor is not, and the energy has to go somewhere.

A9. Correct answer: (A) $u(t) - u(t - T)$

- (A) Correct — the difference of two steps turns a pulse response into a superposition of two step responses.
- (B) Wrong - the sum produces a staircase rising to 2, not a pulse returning to zero.
- (C) Wrong - this scales a step by T and never turns off.
- (D) Wrong - the product is simply $u(t - T)$, which is a delayed step rather than a pulse.

A10. Correct answer: (A) $x(t) = x(\infty) + [x(0+) - x(\infty)] \cdot e^{(-t/\tau)}$

- (A) Correct — the natural-response amplitude is the difference between the initial and final values.
- (B) Wrong - this is the source-free case only and gives the wrong answer whenever the final value is non-zero.
- (C) Wrong - this decays away from the final value rather than toward it.
- (D) Wrong - this has the initial and final values in the wrong roles and does not satisfy either boundary condition.

A11. Correct answer: (A) 2.53 A

- (A) Correct — $\tau = 0.4$ s, so $i = 4(1 - e^{(-1)}) = 4 \times 0.632 = 2.53$ A.
- (B) Wrong - 4 A is the final value, reached only asymptotically.
- (C) Wrong - 1.47 A would be 36.8 percent of the final value, which is how much is still missing, not how much is present.
- (D) Wrong - 0 A is the initial value, correct only at $t = 0$.

A12. Correct answer: (A) $0.693 \cdot RC$

- (A) Correct — $t = -RC \cdot \ln(1 - 0.5) = 0.693 \cdot RC$.
- (B) Wrong - one time constant corresponds to a 63.2 percent threshold, not 50 percent.
- (C) Wrong - 2.3 RC corresponds to a 90 percent threshold.
- (D) Wrong - the delay is not linearly proportional to the threshold fraction; the relation is logarithmic.

A13. Correct answer: (A) 8.65 V

- (A) Correct — at 2τ the response has reached $10(1 - e^{(-2)}) = 8.65$ V.
- (B) Wrong - the asymptote is never reached; using it overstates every later value by 16 percent.
- (C) Wrong - 6.32 V is the value at one time constant, not two.
- (D) Wrong - 3.68 V is what remains of a decay after one time constant, not a charging value.

A14. Correct answer: (A) $\alpha = 1/(2RC)$

- (A) Correct — $1/(2RC)$ is the parallel form; $R/(2L)$ is its series dual.
- (B) Wrong - that is the series formula, and using it for a parallel circuit can flip the damping classification.
- (C) Wrong - that is the resonant frequency ω_0 , not α .
- (D) Wrong - this is dimensionally wrong and increases with R, which is backwards for a parallel circuit.

A15. Correct answer: (A) underdamped

- (A) Correct — $\alpha < \omega_0$ gives complex roots and a ringing response.
- (B) Wrong - overdamped requires α greater than ω_0 .
- (C) Wrong - critical damping requires the two to be exactly equal.
- (D) Wrong - a passive RLC circuit with positive R always decays and is therefore stable.

A16. Correct answer: (A) $(A_1 + A_2t) \cdot e^{(-\alpha t)}$

- (A) Correct — the repeated root needs the extra t factor to supply a second independent solution.
- (B) Wrong - a single exponential has only one constant and cannot satisfy two initial conditions.
- (C) Wrong - that is the overdamped form, which requires two distinct real roots.
- (D) Wrong - that is the underdamped form, which requires complex roots.

A17. Correct answer: (A) increases the damping

- (A) Correct — $\alpha = R/(2L)$ rises with R , so the response becomes less oscillatory.
- (B) Wrong - that is the parallel case, where R is a bypass rather than in the current path.
- (C) Wrong - R appears directly in α for a series circuit, so it cannot be neutral.
- (D) Wrong - ω_0 depends only on L and C and is unaffected by R in either topology.

A18. Correct answer: (A) Two

- (A) Correct — two constants appear in the natural response, so two conditions are required.
- (B) Wrong - one condition leaves the second constant undetermined, and guessing it as zero is usually incorrect.
- (C) Wrong - the order of the equation sets the count, and this circuit is second order.
- (D) Wrong - four conditions would over-determine the system.

A19. Correct answer: (A) a voltage zero crossing

- (A) Correct — at a voltage zero the forced current is at its peak, so the natural term must cancel a large mismatch.
- (B) Wrong - switching at a voltage peak gives nearly the smallest transient in an inductive circuit.
- (C) Wrong - the current is zero before switching, so there is no current peak to switch at.
- (D) Wrong - the offset depends strongly on the switching angle, which is why inrush limiters exist.

A20. Correct answer: (A) a steady sinusoid found by phasor analysis

- (A) Correct — the transient decays and leaves the forced sinusoidal response behind.
- (B) Wrong - zero is the final value only for a source-free circuit.
- (C) Wrong - a constant final value is the DC-excitation case.
- (D) Wrong - the steady state is perfectly well defined and is exactly what phasor analysis computes.

Part B — model answers and marking scheme

B1 (3 marks) — CO3, L2, Ch.7, pp.217-235

Question. State the two continuity rules for switching and justify each from the corresponding element law.

Model answer. Capacitor voltage is continuous: $i = C \cdot dv/dt$, so a step in v would demand infinite current. Inductor current is continuous: $v = L \cdot di/dt$, so a step in i would demand infinite voltage. These are the only two quantities that carry across a switching instant; every other voltage and current may jump and must be recomputed at $t = 0+$ from the new topology. The inductor rule is visible physically as the arc across an opening switch.

Point required	Marks
Both rules stated correctly	1

Point required	Marks
Both justified from the element laws	2
Total	3

B2 (3 marks) — CO3, L3, Ch.8, pp.261-268

Question. Describe the complete procedure for evaluating initial conditions at $t = 0+$ in a second-order circuit.

Model answer. Solve the circuit at $t = 0-$ in DC steady state to obtain $v_C(0-)$ and $i_L(0-)$. Apply continuity to carry these to $t = 0+$. Redraw at $t = 0+$ with the new switch position, capacitors as voltage sources of value $v_C(0+)$ and inductors as current sources of value $i_L(0+)$, then solve that resistive circuit for every other quantity. Finally obtain the derivatives from the element laws: $dv_C/dt(0+) = i_C(0+)/C$ and $di_L/dt(0+) = v_L(0+)/L$.

Point required	Marks
Correct $t = 0-$ solve and continuity step	1
Correct $t = 0+$ redraw with initial-condition sources	1
Correct derivative evaluation	1
Total	3

B3 (3 marks) — CO3, L3, Ch.8, pp.261-268

Question. A 20 V source with 5 Ω and a 2 H inductor has been closed for a long time. At $t = 0$ a 15 Ω resistor is switched into the loop. Find $i(0+)$, $v_L(0+)$ and $di/dt(0+)$.

Model answer. At $t = 0-$ the inductor is a short in DC, so $i_L(0-) = 20/5 = 4$ A. By continuity $i(0+) = 4$ A. At $t = 0+$ the loop resistance is $5 + 15 = 20$ Ω carrying 4 A, so KVL gives $v_L(0+) = 20 - 4 \times 20 = -60$ V. Then $di/dt(0+) = v_L(0+)/L = -60/2 = -30$ A/s. The current is continuous while the inductor voltage jumps.

Point required	Marks
Correct $i(0-)$ and continuity step	1
Correct $v_L(0+)$ from KVL on the new topology	1
Correct derivative	1
Total	3

B4 (3 marks) — CO3, L4, Ch.9, pp.321-334

Question. A parallel RLC has $R = 500$ Ω , $L = 10$ mH and $C = 1$ μ F. Classify the damping and find the damped natural frequency.

Model answer. For a parallel circuit $\alpha = 1/(2RC) = 1/(2 \times 500 \times 1e-6) = 1000$ s^{-1} , and $\omega_0 = 1/\sqrt{LC} = 1/\sqrt{1e-2 \times 1e-6} = 10000$ rad/s. Since $\alpha < \omega_0$ the response is underdamped, with complex characteristic roots. The damped natural frequency is $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)} = \sqrt{(10^8 - 10^6)} = 9950$ rad/s, and the oscillation sits inside an envelope $e^{(-1000t)}$.

Point required	Marks
Correct α using the parallel formula	1
Correct ω_0 and classification	1
Correct ω_d	1
Total	3

B5 (2 marks) — CO3, L4, Ch.9, pp.334-345

Question. Explain why the critically damped response requires a $t \cdot e^{(-\alpha t)}$ term.

Model answer. When $\alpha = \omega_0$ the characteristic equation has a repeated root, so $e^{(-\alpha t)}$ supplies only one independent solution. A second-order equation needs two independent solutions to match two independent initial conditions, and the standard second solution for a repeated root is $t \cdot e^{(-\alpha t)}$. Omitting it leaves a single constant, which can satisfy the initial value or the initial derivative but not both.

Point required	Marks
Repeated-root argument stated	1
Link to needing two independent initial conditions	1
Total	2

B6 (2 marks) — CO3, L4, Ch.9, pp.345-359

Question. Contrast the effect of increasing R in a series RLC with its effect in a parallel RLC.

Model answer. In a series circuit $\alpha = R/(2L)$, so increasing R increases α and therefore increases damping — the response becomes less oscillatory. In a parallel circuit $\alpha = 1/(2RC)$, so increasing R decreases α and decreases damping — the response becomes more oscillatory. The reason is positional: in series the resistance lies in the current path and dissipates the circulating energy, while in parallel it is a bypass whose conductance sets the loss.

Point required	Marks
Both formulas stated with the opposite dependence identified	1
Correct physical explanation of the reversal	1
Total	2

B7 (2 marks) — CO3, L3, Ch.8, pp.272-275

Question. A 10 μF capacitor charged to 5 V discharges through 1 k Ω . Find τ , the initial current, and the total energy dissipated.

Model answer. $\tau = RC = 1000 \times 10\text{e-}6 = 10$ ms. By continuity $v(0+) = 5$ V, so $i(0+) = 5/1000 = 5$ mA. The total energy dissipated over all time equals the energy initially stored, $\frac{1}{2}Cv(0)^2 = \frac{1}{2} \times 10\text{e-}6 \times 25 = 125$ μJ . That total is independent of R ; R sets only the rate at which the energy leaves.

Point required	Marks
Correct τ and initial current	1
Correct energy with the statement of R -independence	1

Point required	Marks
Total	2

B8 (2 marks) — CO3, L3, Ch.8, pp.268-272

Question. Explain how to read a time constant from an oscilloscope trace by two different methods.

Model answer. First, draw the tangent to the decay at $t = 0$; it meets the time axis exactly at $t = \tau$. Second, find the time at which the trace has fallen to 36.8 % of its initial value; that interval is one time constant. A third, more noise-robust method is to replot the data on a logarithmic vertical axis, where the decay becomes a straight line of slope $-1/\tau$.

Point required	Marks
One correct method described	1
A second correct method described	1
Total	2

B9 (2 marks) — CO3, L3, Ch.8, pp.282-286

Question. A 5 V pulse of width 3 ms is applied to an initially uncharged RC circuit with $\tau = 1$ ms. Sketch and describe the capacitor voltage.

Model answer. During the pulse the capacitor charges as $v = 5(1 - e^{-(t/1 \text{ ms})})$, reaching $5(1 - e^{-3}) = 4.75$ V at $t = 3$ ms rather than the full 5 V. When the pulse ends the capacitor discharges from that actual value: $v = 4.75 \cdot e^{-(t-3 \text{ ms})/1 \text{ ms}}$. The response is therefore a rising exponential followed by a falling one, handed over at 4.75 V.

Point required	Marks
Correct charging expression and end-of-pulse value	1
Correct discharge starting from the actual value	1
Total	2

B10 (2 marks) — CO3, L5, Ch.8, pp.295-300

Question. Design an RC power-ON delay of 100 ms for a 5 V supply with a 2.5 V logic threshold, using a 10 μF capacitor.

Model answer. The delay satisfies $t = -RC \ln(1 - V_{\text{th}}/V_s) = -RC \ln(0.5) = 0.693 \cdot RC$. Setting this to 0.1 s gives $RC = 0.1443$ s, so with $C = 10 \mu\text{F}$, $R = 14.43 \text{ k}\Omega$. The nearest standard value is 15 $\text{k}\Omega$, which gives an actual delay of $0.693 \times 15000 \times 10^{-6} = 104$ ms, a 4 % error from component rounding before tolerance is considered.

Point required	Marks
Correct delay formula and RC value	1
Correct R with the standard-value check	1
Total	2

B11 (2 marks) — CO3, L3, Ch.10, pp.371-406

Question. Explain why the final value of a circuit driven by a sinusoid is not zero, and how it is found.

Model answer. The natural response decays to zero, but the forced response persists as long as the source does. Under sinusoidal excitation the forced response is itself a sinusoid at the source frequency, so the circuit settles into a continuing AC steady state rather than a constant. It is found by phasor analysis: convert every element to an impedance at the source frequency, solve, and convert the answer back to the time domain.

Point required	Marks
Correct distinction between natural decay and persistent forced response	1
Correct phasor procedure for finding it	1
Total	2

B12 (2 marks) — CO3, L4, Ch.8, pp.286-295

Question. Explain why a 20 V source switched onto a series 5 Ω and 2 H gives $i(t) = 4(1 - e^{(-2.5t)})$ A, identifying each part of the response.

Model answer. The forced response is the DC final value, $i(\infty) = 20/5 = 4$ A, obtained with the inductor shorted. The natural response is $Ae^{(-t/\tau)}$ with $\tau = L/R = 0.4$ s, so the exponent is $-2.5t$. The complete response is $i(t) = 4 + Ae^{(-2.5t)}$. Applying $i(0+) = 0$ to the complete expression gives $A = 0 - 4 = -4$, hence $i(t) = 4(1 - e^{(-2.5t)})$ A.

Point required	Marks
Forced response and τ identified correctly	1
Constant found by applying the initial condition to the complete expression	1
Total	2

B13 (2 marks) — CO3, L5, Ch.9, pp.359-361

Question. Explain why switching an inductive load onto an AC supply at a voltage zero produces the worst transient.

Model answer. The forced current lags the voltage, so at a voltage zero crossing the steady-state current is near its peak. Continuity requires the actual current to start from zero, so the natural response must supply a large offset to cancel that mismatch, and the first peak can approach twice the steady-state peak. Switching at a voltage peak puts the forced current near zero, so almost no offset is needed. This is the mechanism behind transformer inrush and why inrush limiters are specified.

Point required	Marks
Correct link between switching angle and the forced current value	1
Correct conclusion about the offset magnitude and its engineering consequence	1
Total	2